Fine-Tuning HuggingFace Embedding Models for Asymmetric Search

Train embedders on your data

Asymmetric Search: Why It Matters

- Short Queries ≠ Long Chunks

QUERY

What days of the week are campus tours offered at Christopher Newport ' 'University?

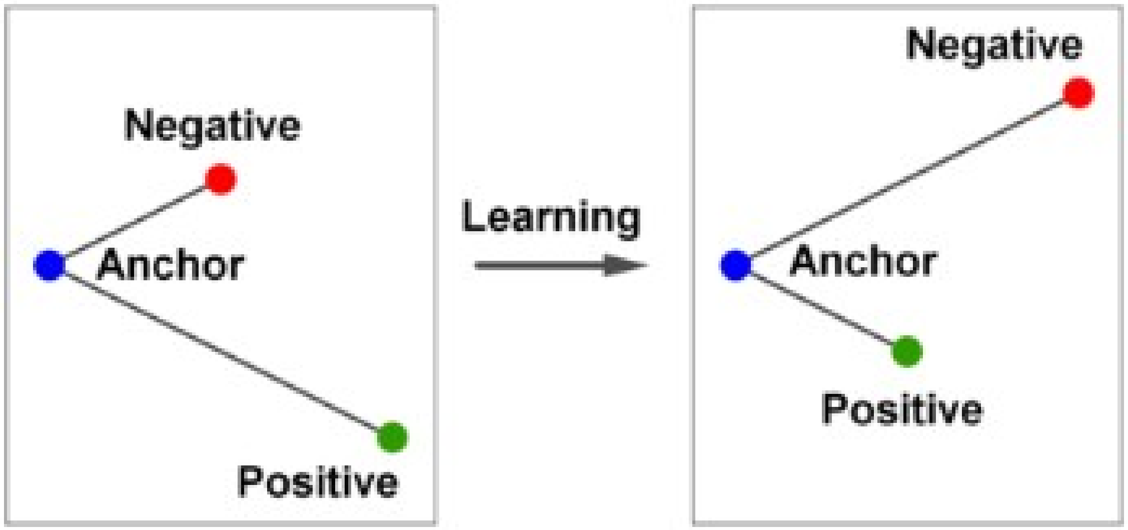
Chunk

'Come See for Yourself \n' 'Ready to picture your life as a CNU Captain? Schedule \n' 'a visit and see why so many students choose to call Chris-\n' 'topher Newport home. \n' 'Campus tours are offered Monday through Saturday \n' '(except major holidays). Visit us online at admission.cnu. \n' 'edu or call (757) 594-7015/(800) 333-4268. \n' 'We look forward to showing you why Christopher New-\n' 'port is the number one regional public university in Virginia!'

- Embedder Model must learn directional alignment
- Goal: Fine tuning improves retrieval accuracy

Loss Functions

$$\mathcal{L} = max(d(a,p) - d(a,n) + margin, 0)$$



Want to push negative away and pull positive closer

source https://en.wikipedia.org/wiki/Triplet_loss#:~:text=Triplet%20loss%20is%20a%20machine,data%20points%20correspond%20to%20images.

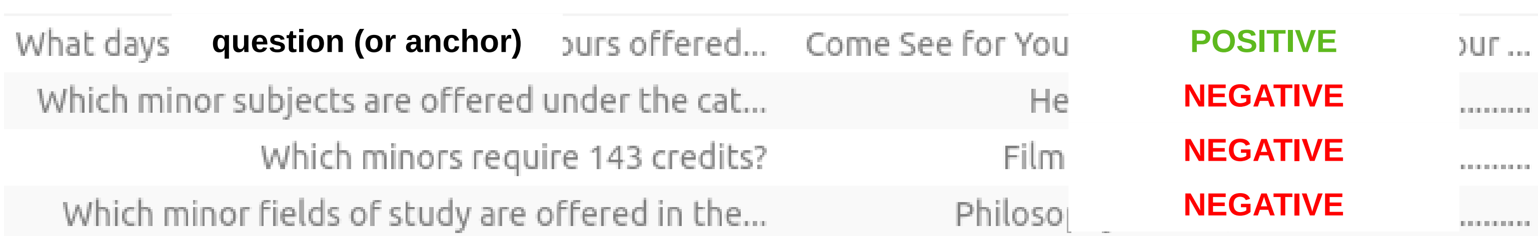
Loss Functions

- MultipleNegativesRankingLoss (MNRL)
- Uses ‘in-batch’ negatives
- ex. For this batch of 4

What days of the week are campus tours offered...	Come See for Yourself \nReady to picture your ...
Which minor subjects are offered under the cat...	Health Studies ....
Which minors require 143 credits?	Film Studies, minor ....
Which minor fields of study are offered in the...	Philosophy of Law, minor ....

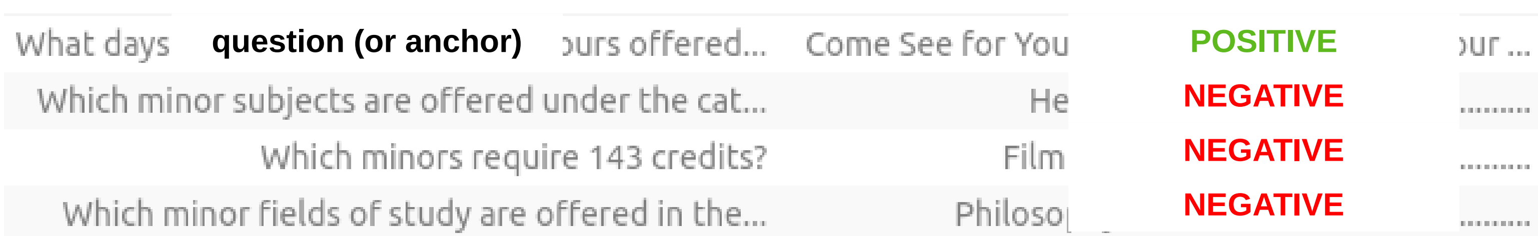
Loss Functions

- MultipleNegativesRankingLoss (MNRL)
- Uses ‘in-batch’ negatives
- ex. For this batch of 4, we get 3 NEGATIVES



Loss Functions

- MultipleNegativesRankingLoss (MNRL)
- Uses ‘in-batch’ negatives
- ex. For this batch of 4, we get 3 NEGATIVES



The larger the batch the more negatives. But they are randomly picked, so easy negatives

Loss Functions

TripletLoss

Uses explicit (anchor, positive, negative)

Loss Functions

TripletLoss

Uses explicit (anchor, positive, negative)

How does it work?

Loss Functions

$$\mathcal{L} = max(d(a,p) - d(a,n) + margin, 0)$$

Margin=1

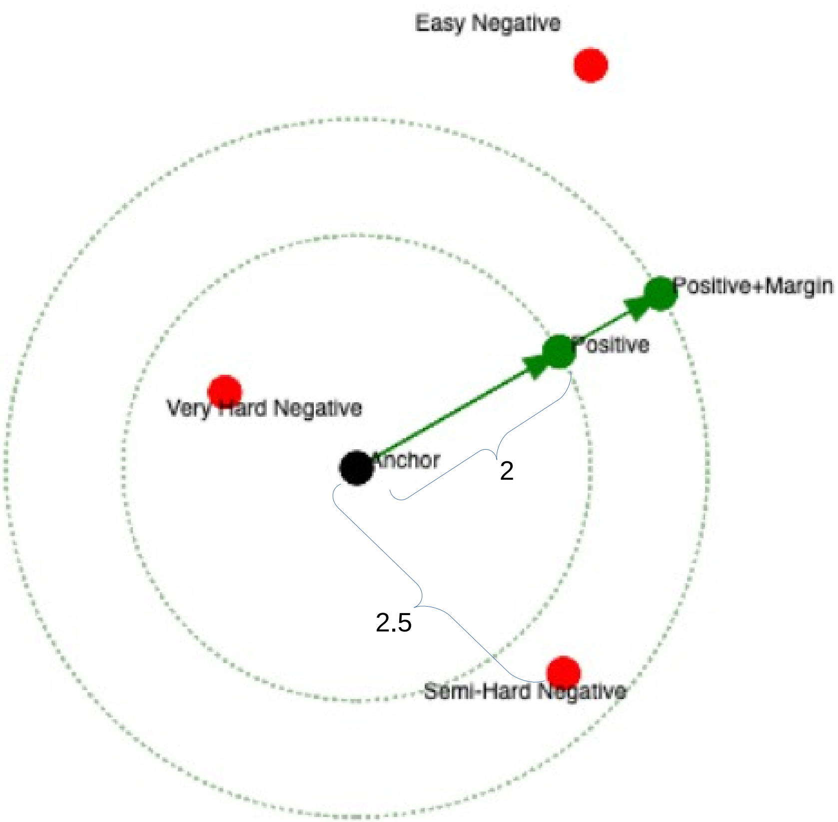
Loss=max(2-4+1,0) => 0

no loss, so nothing to learn here

source https://en.wikipedia.org/wiki/Triplet_loss#:~:text=Triplet%20loss%20is%20a%20machine,data%20points%20correspond%20to%20images.

Loss Functions

$$\mathcal{L} = max(d(a,p) - d(a,n) + margin, 0)$$



Margin=1

Loss=max(2-2.5+1,0) => .5

.5 loss learn from this one!

source https://en.wikipedia.org/wiki/Triplet_loss#:~:text=Triplet%20loss%20is%20a%20machine,data%20points%20correspond%20to%20images.

Loss Functions

$$\mathcal{L} = max(d(a,p) - d(a,n) + margin, 0)$$

Margin=1

Loss=max(2-3+1,0) => 0 no learning from this one

source https://en.wikipedia.org/wiki/Triplet_loss#:~:text=Triplet%20loss%20is%20a%20machine,data%20points%20correspond%20to%20images.

Loss Functions

$$\mathcal{L} = max(d(a,p) - d(a,n) + margin, 0)$$

- Want ‘hard’ negatives
- Not ‘easy’ negatives
- Want negatives that are ‘close’

source https://en.wikipedia.org/wiki/Triplet_loss#:~:text=Triplet%20loss%20is%20a%20machine,data%20points%20correspond%20to%20images.

Loss Functions

$$\mathcal{L} = max(d(a,p) - d(a,n) + margin, 0)$$

- Want ‘hard’ negatives
- Not ‘easy’ negatives
- Want negatives that are ‘close’
- How?

source https://en.wikipedia.org/wiki/Triplet_loss#:~:text=Triplet%20loss%20is%20a%20machine,data%20points%20correspond%20to%20images.

Loss Functions

$$\mathcal{L} = max(d(a,p) - d(a,n) + margin, 0)$$

- Want 'hard' negatives
- Not 'easy' negatives
- Want negatives that are 'close'
- How?
- Pull from Vector DB.

source https://en.wikipedia.org/wiki/Triplet_loss#:~:text=Triplet%20loss%20is%20a%20machine,data%20points%20correspond%20to%20images.

Hard Negative (HN) Mining

- Encode all chunks → store in ChromaDB
- Encode question → query top-K closest chunks
- Filter true positive → keep top 3–5 as Hns

Hard Negative (HN) Mining

- Encode all chunks → store in ChromaDB
- Encode question → query top-K closest chunks
- Filter true positive → keep top 3–5 as Hns

Original Anchor	**Orig Positive**

Start with this

Hard Negative (HN) Mining

- Encode all chunks → store in ChromaDB
- Encode question → query top-K closest chunks
- Filter true positive → keep top 3–5 as Hns

Original Anchor	**Orig Positive**	**Hard Negative 1**
Original Anchor	**Orig Positive**	**Hard Negative 2**
Original Anchor	**Orig Positive**	**Hard Negative 3**
Original Anchor	**Orig Positive**	**Hard Negative 4**

Finish with this

Hard Negative (HN) Mining

- Encode all chunks → store in ChromaDB
- Encode question → query top-K closest chunks
- Filter true positive → keep top 3–5 as Hns
- This dramatically expands training dataset

Original Anchor	**Orig Positive**	**Hard Negative 1**
Original Anchor	**Orig Positive**	**Hard Negative 2**
Original Anchor	**Orig Positive**	**Hard Negative 3**
Original Anchor	**Orig Positive**	**Hard Negative 4**

New rows

New Column

2-Stage Embedder Fine-Tuning

- Stage 1: Fine tune model using MNRL on
  (question, chunk) pairs
- Embed corpus using fine tuned model
- Hard Negative Mine to expand training dataset
- Stage 2: Fine tune model using TripletLoss on
  (question, positive, hard negative)

Key Takeaways

- Hard negatives dramatically improve model quality
- Multi-stage finetuning most effective